

Continuous Dynamical Systems for Weighted Bipolar Argumentation

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Abstract

Weighted bipolar argumentation frameworks determine the strength of arguments based on an initial weight and the strength of their attackers and supporters. These frameworks can be applied to model and solve problems that arise in areas like social media analysis and decision support. Approaches for computing strength values often assume an acyclic argumentation graph and successively set arguments' strength based on the strength of their parents. A natural idea for cyclic graphs is to update strength values simultaneously rather than successively. Continuous dynamical systems seem well-suited for this approach because they can feature better convergence behaviour than their discrete counterparts. We investigate such a system here. For acyclic graphs, our model is guaranteed to converge and solutions can be computed as efficiently as for successive update procedures. We currently cannot prove very general guarantees for cyclic frameworks, but give empirical evidence that our model converges quickly even in complex cyclic graphs with thousands of nodes and ten thousands of edges. We also explain how potential oscillations can be detected visually. Our model's axiomatic properties complement existing approaches. We also give sufficient conditions under which successive update procedures can be transformed to well-defined dynamical systems with similar guarantees easily.

1 Introduction

Abstract argumentation (Dung 1995) studies the acceptability of arguments based purely on their relationships and abstracted from their content. Whereas in the classical framework arguments can only be attacked by other arguments, they can also be supported in *bipolar argumentation* (Amgoud, Cayrol, and Lagasquie-Schiex 2004; Oren and Norman 2008; Cayrol and Lagasquie-Schiex 2013; Polberg and Oren 2014). The classical approach is to decide which arguments can be accepted according to different semantics (Baroni, Caminada, and Giacomin 2011). However, recent experiments show that a more fine-grained view is beneficial (Polberg and Hunter 2018). Quantifying arguments by probabilistic degrees of belief is one natural choice (Thimm 2012; Hunter 2013; Hunter and Potyka 2017), but one may also apply tools from game theory (Baroni et al. 2017) or rank arguments. Rankings can be obtained

from numerical values obtained from fixed point equations (Besnard and Hunter 2001; Leite and Martins 2011; Correia, Cruz, and Leite 2014; Barringer, Gabbay, and Woods 2012) or directly from the graph structure (Cayrol and Lagasquie-Schiex 2005; Amgoud and Ben-Naim 2013). In order to evaluate such approaches, various axioms have been proposed (Amgoud and Ben-Naim 2016a; 2016b; Bonzon et al. 2016; Hunter and Thimm 2017).

Recently some weighted bipolar argumentation frameworks have been introduced that start with an initial weight of arguments and adapt this weight based on the strength of attacking and supporting arguments (Baroni et al. 2015; Rago et al. 2016; Budán, Simari, and Simari 2016; Amgoud and Ben-Naim 2017). For instance, when analyzing discussions in social networks, the initial weight can be based on the number of likes, dislikes and shares of a post, and attack and support relations can be identified from replies using sentiment analysis tools (Leite and Martins 2011; Alsinet et al. 2017). Another natural application is decision support in areas like engineering design and e-democracy (Baroni et al. 2013; Rago et al. 2016). In this context, arguments may correspond to issues that have to be decided and pro and contra arguments that support and attack issues or other arguments. The initial weight may correspond to an initial bias towards accepting or rejecting an argument or may be based on the reputation of the argument's source.

When the argumentation graph is acyclic, the final strength of arguments can be computed successively according to a topological ordering. However, for cyclic graphs, computing strength values is less straight-forward. One natural idea to deal with cycles is to update arguments simultaneously rather than successively. A continuous dynamical systems approach seems well suited for this idea. At each point in time, the strength of arguments evolves continuously based on the strength of attackers and supporters. If this evolutionary process converges to an equilibrium state, we can use the strength values in equilibrium to define the final strength of arguments. In the following, we will investigate the *Quadratic Energy Model* that is based on such a system.

2 Background

Weighted bipolar argumentation frameworks have been studied in (Baroni et al. 2015; Rago et al. 2016; Amgoud and

Ben-Naim 2017). One motivation for the *QuAD-algorithm* from (Baroni et al. 2015) was to evaluate the strength of answers in decision-support systems. While the algorithm computes reasonable results, the authors also noted that it can behave discontinuously. For instance, an unsupported argument with strength 0 can jump from 0 to a comparatively high value even when adding only a very weak supporter (Rago et al. 2016). This can be undesirable in some applications. The *discontinuity-free QuAD-algorithm DF-QuAD* from (Rago et al. 2016) avoids this behaviour.

A collection of axioms for evaluating weighted bipolar argumentation frameworks has been introduced in (Amgoud and Ben-Naim 2017). The authors noted that the approaches in (Baroni et al. 2015; Rago et al. 2016) can violate some of them. We will discuss the axioms in Section 4. For now, let us just focus on one reason why they can be violated and why this may be undesirable. Both QuAD algorithms aggregate the strength of multiple attackers and supporters in a geometric fashion by multiplying them in a suitable way. A consequence of this definition is that a single attacker (supporter) with strength 1 *saturates* the aggregated attacking (supporting) strength. It does not matter whether there is one single or one-hundred arguments with strength 1. If an argument has only one supporter and one attacker with strength 1, this is a reasonable aggregate. However, if there is only one attacker with strength 1, but two supporters with strength 1, one could argue that the aggregated support value should be larger than the aggregated attack value. However, the *saturation property* of the QuAD algorithms makes this impossible. Because of this, some intuitive axioms from (Amgoud and Ben-Naim 2017) can be violated.

The *Euler-based restricted semantics* from (Amgoud and Ben-Naim 2017) avoids the saturation property by aggregating the strength of arguments in a linear fashion. The *energy* at an argument is defined as $E = \sum_{i \in \text{Sup}} s_i - \sum_{i \in \text{Att}} s_i$ where Sup and Att are the sets of indices of supporters and attackers and the s_i are the corresponding strength values. The aggregated strength at an argument is then defined as

$$1 - \frac{1 - w^2}{1 + w \cdot \exp(E)}, \quad (1)$$

where w is the initial weight of the argument. The Euler-based restricted semantics satisfies the axioms proposed in (Amgoud and Ben-Naim 2017), but only if the initial weights 0 and 1 are excluded as we explain in Section 4.

Another problem of the Euler-based restricted semantics is that the aggregation formula (1) is defined in an asymmetric fashion. This causes an unbalance between attack and support. Consider two arguments A, B both with initial weight 0.5. Suppose there is only an edge from A to B . If the edge is attacking, our final strength for B will be $1 - \frac{1 - 0.5^2}{1 + 0.5 \cdot \exp(-0.5)} \approx 0.425$. If the edge is supporting, we will have $1 - \frac{1 - 0.5^2}{1 + 0.5 \cdot \exp(0.5)} \approx 0.589$. That is, the support increases the strength by about 0.089, whereas the attack decreases the strength by only about 0.075. Hence, the effect of support is about 15% stronger in this example. However, arguably, both attack and support should move the strength away from 0.5 by the same amount.

We will now introduce a model that aggregates arguments with strength 1 in a more fine-grained way than the QuAD algorithms while avoiding the asymmetric behaviour of the Euler-based restricted semantics. In particular, strength values will be updated simultaneously and continuously rather than successively in discrete time steps.

3 The Quadratic Energy Model

We consider *weighted bipolar argumentation graphs (BAGs)* as introduced in (Amgoud and Ben-Naim 2017).

Definition 1 (BAG). A BAG is a quadruple $\mathbf{A} = (\mathcal{A}, w, \mathcal{R}, \mathcal{S})$, where $\mathcal{A}$ is a finite set of arguments, $w : \mathcal{A} \rightarrow [0, 1]$ is a weight function and $\mathcal{R}$ and $\mathcal{S}$ are binary relations on $\mathcal{A}$ called *attack* and *support*.

In order to simplify notation, we assume that the i -th argument is called i , that is, $\mathcal{A} = \{1, \dots, n\}$. The weight function w defines an initial strength value between 0 and 1 for each argument. If aRb (aSb), we say that a attacks (supports) b . We let $\text{Att}_i = \{h \in \mathcal{A} \mid hRi\}$ denote i 's attackers and let $\text{Sup}_i = \{h \in \mathcal{A} \mid hSi\}$ denote i 's supporters.

Roughly speaking, our goal is to find a vector $s^* \in \mathbb{R}^n$ that assigns a final strength value s_i^* to each argument i based on i 's initial weight and the strength of i 's attackers and supporters. We will do this in an indirect way. First, we try to find a function $s : \mathbb{R}_0^+ \rightarrow \mathbb{R}^n$ that maps non-negative time points to strength vectors. Intuitively, $s_i(t)$ is the strength of argument i at time t . In particular, we want $s_i(0) = w(i)$. If all $s_i(t)$ converge to an equilibrium state s_i^* as time progresses, we let $s^* = \lim_{t \rightarrow \infty} s(t)$.

Before thinking about the existence of the limit, let us focus on finding an appropriate function s . We assume that s is a continuously differentiable function. Intuitively, this assumption guarantees that the strength values change smoothly as time progresses. In particular, it allows us to describe s by a system of differential equations. We will explain the high-level idea of all building blocks to make the paper comprehensible for readers who are unfamiliar with differential equations. For more details, we refer to textbooks like (Polyanin and Zaitsev 2017).

We will describe s by n nonlinear but continuous differential equations that define the derivative $\frac{ds_i}{dt}$ for all $i \in \mathcal{A}$. Intuitively, $\frac{ds_i}{dt}$ describes the continuous change at argument i based on the current strength of its attackers and supporters. The column vector consisting of the derivatives $\frac{ds_i}{dt}$ is called the *Jacobian matrix* of s . We denote the Jacobian matrix by $\frac{ds}{dt}$, that is, $\frac{ds}{dt}$ is an n -dimensional column vector whose i -th component is $\frac{ds_i}{dt}$. The Jacobian matrix at time t describes the current change of s at time t .

We can design our differential equations such that the limit $s^* = \lim_{t \rightarrow \infty} s(t)$ satisfies desirable properties if it exists. Some important properties that guided the design of the following system are the following:

1. If the energy at i is 0 in the limit, then $s_i^* = w(i)$.
2. Every attacker (supporter) with positive strength should push the strength of i closer to 0 (1).
3. The impact of positive and negative energy should be symmetrical.

We now present our system of differential equations and show that it uniquely defines a strength model for every BAG. In Section 4, we will prove that it does indeed have the desired behaviour.

Definition 2 (The Quadratic Energy Model). Let $\mathbf{A}$ be a BAG. For all $j \in \mathcal{A}$, the *energy at j* is defined as

$$E_j = \sum_{i \in \text{Sup}_j} s_i - \sum_{i \in \text{Att}_j} s_i$$

and for all $x \in \mathbb{R}$, the *impact of x* is defined as

$$h(x) = \frac{\max\{x, 0\}^2}{1 + \max\{x, 0\}^2}.$$

The *quadratic energy model* $\sigma^{\mathbf{A}} : \mathbb{R}_0^+ \rightarrow \mathbb{R}^n$ for $\mathbf{A}$ is the unique solution of the system of differential equations

$$\begin{aligned} \frac{ds_j}{dt} = & w(j) - s_j + (1 - w(j)) \cdot h(E_j) \\ & - w(j) \cdot h(-E_j), \quad j \in \mathcal{A} \end{aligned} \quad (2)$$

with initial conditions $s_j(0) = w(j)$.

Proposition 1 (Well-Definedness). *The system (2) with the initial conditions $s_j(0) = w(j)$ for $j = 1, \dots, n$ has a unique solution $\sigma^{\mathbf{A}} : \mathbb{R}_0^+ \rightarrow \mathbb{R}^n$.*

Proof. We show that the right-hand side of (2) is continuously differentiable everywhere. The claim follows then from existence and uniqueness theorems for nonlinear systems of ordinary differential equations stated in (Polyanin and Zaitsev 2017, Section 7.1.2). $\max\{x, 0\}^2$ is differentiable everywhere with continuous derivative $\max\{2x, 0\}$. Linearity of the derivative and the quotient rule imply that the impact h is continuously differentiable everywhere (in particular, the denominator in the derivative is always greater than 1). The chain rule then implies that $h(E_j)$ and $h(-E_j)$ are continuously differentiable everywhere with respect to all s_i . Finally, linearity of derivatives implies that the right-hand side of (2) is continuously differentiable everywhere. $\square$

Remark 1. Let us note a few things about Definition 2.

1. The impact $h(x)$ is 0 for all $x \leq 0$ and then increases strictly monotonically. In particular, we have $h(E_j) = h(-E_j) = 0$ if and only if $E_j = 0$ and either $h(E_j) = 0$ or $h(-E_j) = 0$ if $E_j \neq 0$. Notice also that $\lim_{x \rightarrow \infty} h(x) = 1$.
2. If $E_j = 0$, (2) becomes $w(j) - s_j$. If $s_j < w(j)$ ($s_j > w(j)$), the difference will be positive (negative) and s_j will increase (decrease). Therefore, the strength of j will be drawn to the initial weight whenever the energy is 0.
3. If $E_j \neq 0$, we have either $(1 - w(j)) \cdot h(E_j) = 0$ or $w(j) \cdot h(-E_j) = 0$. For instance, if $E_j > 0$, (2) becomes $w(j) - s_j + (1 - w(j)) \cdot h(E_j)$. As E_j goes to infinity, this term will converge to $w(j) - s_j + (1 - w(j)) = 1 - s_j$. s_j will then tend towards 1 for similar reasons as in item 2. In general, if $E_j > 0$, the term $(1 - w(j)) \cdot h(E_j)$ will push the strength to a value between $w(j)$ and 1. Symmetrically, if $E_j < 0$, the term $-w(j) \cdot h(-E_j)$ will push the strength to a value between $w(j)$ and 0.

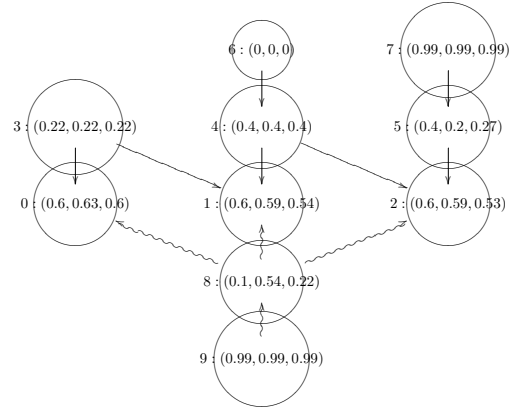


Figure 1: BAG from (Amgoud and Ben-Naim 2017). Nodes show (initial weight, strength under quadratic energy model, strength under Euler-based restricted semantics).

Proposition 1 tells us that there exists a unique function $\sigma^{\mathbf{A}}$ that features our desired dynamic behaviour. Unfortunately, simple closed-form solutions for nonlinear dynamical systems rarely exist. However, $\sigma^{\mathbf{A}}$ can be approximated numerically. To understand how, recall that the derivative at time t describes the change of strength at time t . Intuitively, this allows us to approximate the solution by starting from the weights at time 0 that are given by the weight function w . We then compute the Jacobian for the current strength values from (2) and shift the strength values by a small step δ in the corresponding direction. This gives us an approximation of the function value at time δ . By repeating this process, we can construct the function value at time $2 \cdot \delta$, $3 \cdot \delta$ and so on. This is basically Euler's method for solving initial value problems. More sophisticated methods like Runge—Kutta methods have stronger approximation guarantees. For instance, for the classical Runge—Kutta method RK4, the approximation error is in the order of $O(\delta^4)$. In practice, this means that if we halve the step size δ (double the number of iterations), we usually decrease the approximation error by a factor of 16 (Polyanin and Zaitsev 2017).

Figure 1 illustrates our approach with a BAG from (Amgoud and Ben-Naim 2017). For each argument, the node shows the initial weight, the strength values that the quadratic energy model converges to and the strength values obtained for the Euler semantics from (Amgoud and Ben-Naim 2017). We denote attacks by straight arrows and supports by squiggly arrows. Note that under the quadratic energy model, the strength of 8 is increased much more significantly by the strong supporter 9 than under the Euler semantics. Similarly, 5's strength decreases more significantly due to the attack by 7. However, both approaches tend to increase and decrease the same arguments in this example.

Let us emphasize again that the quadratic energy model is a function $\sigma^{\mathbf{A}} : \mathbb{R}_0^+ \rightarrow \mathbb{R}^n$ and that the strength values are obtained from the limit $s^* = \lim_{t \rightarrow \infty} \sigma^{\mathbf{A}}(t)$ provided that the limit exists. Figure 2 shows the long-term behaviour of $\sigma^{\mathbf{A}}(t)$. $\sigma^{\mathbf{A}}(t)$ visually converges at about time 5.5. The figure shows the function graphs of $\sigma_i^{\mathbf{A}}(t)$ corresponding to the strength of argument i . For instance, the green graph

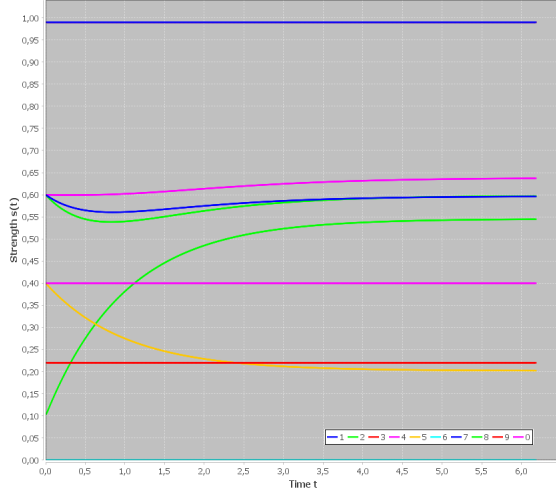


Figure 2: Long-term behaviour of σ^A for BAG in Figure 1.

rising from 0.1 to 0.54 corresponds to the strength of 8. The green graph starting from 0.6 corresponds to the strength of 2. Notice that 2's strength initially decreases due to the attacks by 4 and 5. As time progresses, 5 becomes weaker and the supporter 8 becomes stronger, so that 2's strength finally increases until it reaches an equilibrium state.

We close this section with two basic guarantees. Let us first note that strength values are always in the desired range.

Lemma 1 (Bounded Strength). $\sigma^A(t) \in [0, 1]$ for all $t \geq 0$.

Proof. Note that the initial conditions in Definition 2 demand that the strength values at time 0 correspond to the initial weights that are in $[0, 1]$. Hence, $\sigma^A(0) \in [0, 1]$. By continuity of $\sigma^A(t)$, we can only leave this range when the function value is 0 or 1. Assume $\sigma^A(t') = 1$. Then equation (2) implies that $\frac{ds_j}{dt}(t) \leq w(j) - 1 + (1 - w(j)) = 0$ for all $j = 1, \dots, n$ because the impact $h(x)$ is a non-negative function that is bounded from above by 1 (notice that the inequality is actually strict whenever $w(j) < 1$). Hence, σ^A is non-increasing at 1 and hence cannot take a value larger than 1. We can argue symmetrically that σ^A can never take a value smaller than 0. $\square$

If the limit $s^* = \lim_{t \rightarrow \infty} \sigma^A(t)$ exists, we call s^* the *equilibrium state reached by σ^A* . The next proposition explains what strength values we can expect in equilibria.

Proposition 2 (Strength in Equilibrium States). *If the limit $s^* = \lim_{t \rightarrow \infty} \sigma^A(t)$ exists, then we have*

$$s_j^* = w(j) + (1 - w(j)) \cdot h(E_j) - w(j) \cdot h(-E_j). \quad (3)$$

Proof. If the limit exists, the Jacobian matrix must converge to the 0-vector, that is, $\lim_{t \rightarrow \infty} \frac{d\sigma^A}{dt}(t) = 0$. The claim follows by setting equation (2) to 0 and solving for s_j . $\square$

Example 1. Consider the equilibrium values for the BAG in Figure 1. For argument 1, the energy is $E_1 = s_8 - s_3 - s_4 \approx 0.54 - 0.22 - 0.4 \approx -0.08$. We have $w(1) = 0.6$ and $s_1 = w(1) - w(1) \cdot h(-0.08) \approx 0.6 - 0.6 \cdot \frac{0.08^2}{1+0.08^2} \approx 0.5962$.

In the next section, we will show that equilibrium states are guaranteed to satisfy our desiderata.

4 Axiomatic Properties

We now turn to axiomatic properties that equilibrium states $s^* = \lim_{t \rightarrow \infty} \sigma^A(t)$ of the quadratic energy model satisfy. To begin with, we show that all axioms from (Amgoud and Ben-Naim 2017) are satisfied. For some properties that are easy to check, but difficult to phrase, we omit a precise definition to meet space restrictions. Please see (Amgoud and Ben-Naim 2017) for more details.

Anonymity (Amgoud and Ben-Naim 2017) states that isomorphic frameworks yield the same strength values. We omit a precise definition because it comes down to demanding that renaming arguments does not change the final strength, which is clear for the quadratic energy model.

Proposition 3 (Anonymity). *If A and A' are isomorphic, then $\lim_{t \rightarrow \infty} \sigma^A(t) = \lim_{t \rightarrow \infty} \sigma^{A'}(t)$.*

Independence (Amgoud and Ben-Naim 2017) says that the strength of arguments is independent of disconnected arguments. This is formalized by taking the union of disjoint BAGs and demanding that the strength values remain unchanged. The weight function that equals w (w') on the domain of w (w'), we denote a little sloppily by $w \cup w'$.

Proposition 4 (Independence). *Let $A = (\mathcal{A}, w, \mathcal{R}, S)$ and $A' = (\mathcal{A}', w', \mathcal{R}', S')$ with $\mathcal{A} = \{1, \dots, n\}$, $\mathcal{A}' = \{n+1, \dots, n+m\}$. Assume A and A' reach equilibria s and s' . Then $A'' = (\{1, \dots, n+m\}, w \cup w', \mathcal{R} \cup \mathcal{R}', S \cup S')$ reaches an equilibrium $s'' = \lim_{t \rightarrow \infty} \sigma^{A''}(t)$ and $s_i'' = s_i$ for $i = 1, \dots, n$ and $s_i'' = s'_{i-n}$ for $i = n+1, \dots, n+m$.*

Proof. This is clear because the differential equations for both BAGs are completely independent. $\square$

Directionality (Amgoud and Ben-Naim 2017) states that the strength of an argument depends only on its predecessors in the graph.

Proposition 5 (Directionality). *Let $A = (\mathcal{A}, w, \mathcal{R}, S)$ and $A' = (\mathcal{A}', w', \mathcal{R}', S')$ with $\mathcal{A} = \mathcal{A}'$, $\mathcal{R} \subseteq \mathcal{R}'$, $S \subseteq S'$, $\mathcal{R} \cup S = \mathcal{R}' \cup S' \cup \{(a, b)\}$. Assume A and A' reach equilibria s and s' . Then for all $i \in \mathcal{A}$ such that there is no directed path from b to i , we have $s_i = s'_i$.*

Proof. Since there is no path from b to i , b cannot be parent of any of i 's predecessors. However, as Proposition 2 explains, the final strength of each argument depends only on the final strength of its parents. Since the subgraph consisting of i 's predecessors in A must equal the subgraph in A' , the strength of all arguments in this subgraph must converge to the same values in both A and A' . $\square$

Equivalence (Amgoud and Ben-Naim 2017) demands that if two arguments have the same weight and equally strong attackers and supporters, then they should have the same strength.

Proposition 6 (Equivalence). *Let $A = (\mathcal{A}, w, \mathcal{R}, S)$ be a BAG that reaches an equilibrium s . Then for all $a, b \in \mathcal{A}$ such that $w(a) = w(b)$ and there are bijections $f : \text{Att}_a \rightarrow$*

$\text{Att}_b, g : \text{Sup}_a \rightarrow \text{Sup}_b$ such that $s_x = s_{f(x)}$ and $s_y = s_{g(y)}$ for all $x \in \text{Att}_a, y \in \text{Sup}_a$, then $s_a = s_b$.

Proof. The claim can be seen from Proposition 2 because $w(a) = w(b)$ and the assumptions imply $E_a = E_b$. $\square$

Stability (Amgoud and Ben-Naim 2017) states that if an argument has neither attackers nor supporters, then its strength should just be its initial weight.

Proposition 7 (Stability). *Let $A = (\mathcal{A}, w, \mathcal{R}, S)$ be a BAG that reaches an equilibrium s . If there is an $a \in \mathcal{A}$ such that $\text{Att}_a = \text{Sup}_a = \emptyset$, then $s_a = w(a)$.*

Proof. This follows from Proposition 2 because the assumptions imply $E_a = 0$. $\square$

Neutrality (Amgoud and Ben-Naim 2017) demands that arguments with strength 0 do not have any impact.

Proposition 8 (Neutrality). *Let $A = (\mathcal{A}, w, \mathcal{R}, S)$ be a BAG that reaches an equilibrium s . If there are $a, b \in \mathcal{A}$ such that $\text{Att}_a \subseteq \text{Att}_b, \text{Sup}_a \subseteq \text{Sup}_b, \text{Att}_a \cup \text{Sup}_a = \text{Att}_b \cup \text{Sup}_b \cup \{d\}$ and $s_d = 0$, then $s_a = s_b$.*

Proof. The claim follows from Proposition 2 because the assumptions imply $E_a = E_b$ (d adds zero). $\square$

Monotony (Amgoud and Ben-Naim 2017) states that if a is equally or weaker attacked than b and equally or stronger supported and both have the same initial weight, then a 's strength must be equal or greater than b 's strength. *Strict Monotony* (Amgoud and Ben-Naim 2017) demands that the strength is strictly greater in certain cases. In order to explain this formally, we will write Att^+ and Sup^+ to denote the subsets of arguments in Att and Sup with strictly positive strength in the given equilibrium. As opposed to the formulation in (Amgoud and Ben-Naim 2017), we add the assumption $w(a), w(b) < 1$ to the axiom because it would otherwise imply that if $w(a) = w(b) = 1, \text{Att}_a = \text{Att}_b = \emptyset$ and $\text{Sup}_a^+ \supset \emptyset = \text{Sup}_b$, then $s_a > s_b$. But then $s_b < 1$ which contradicts *Stability*. Indeed, we should expect $s_a = s_b = 1$ in this case because there is no reason to weaken b .

Proposition 9 (Monotony). *Let $A = (\mathcal{A}, w, \mathcal{R}, S)$ be a BAG that reaches an equilibrium s . If $a, b \in \mathcal{A}$ are such that $0 < w(a) = w(b) < 1, \text{Att}_a \subseteq \text{Att}_b, \text{Sup}_a \supseteq \text{Sup}_b$, then*

1. $s_a \geq s_b$. (Monotony)
2. if furthermore $(s_a > 0 \text{ or } s_b < 1)$ and $(\text{Att}_a^+ \subset \text{Att}_b^+ \text{ or } \text{Sup}_a^+ \supset \text{Sup}_b^+)$, then $s_a > s_b$. (Strict Monotony)

Proof. Both claims follows from Proposition 2 because $E_a \geq E_b$ ($E_a > E_b$) when the condition of (strict) monotony is satisfied and $w(a) = w(b)$. $\square$

Monotony makes sure that the quantity of attackers and supporters of an argument has a reasonable impact on its final strength value. *Reinforcement* (Amgoud and Ben-Naim 2017) gives a similar guarantee for their quality. Again there is a strict version. For similar reasons as before, we add the assumption $w(a), w(b) < 1$ to the axiom.

Proposition 10 (Reinforcement). *Let $A = (\mathcal{A}, w, \mathcal{R}, S)$ be a BAG that reaches an equilibrium s . If $a, b \in \mathcal{A}$ are such that $0 < w(a) = w(b) < 1, \text{Att}_a \setminus \{x\} = \text{Att}_b \setminus \{y\}, \text{Sup}_a \setminus \{x'\} = \text{Sup}_b \setminus \{y'\}, s_x \leq s_y$ and $s_{x'} \geq s_{y'}$, then*

1. $s_a \geq s_b$. (Reinforcement)
2. if $(s_a > 0 \text{ or } s_b < 1)$ and $(s_x < s_y \text{ or } s_{x'} > s_{y'})$, then $s_a > s_b$. (Strict Reinforcement)

Proof. Both claims follows from Proposition 2 because $E_a \geq E_b$ ($E_a > E_b$) when the condition of (strict) reinforcement is satisfied and $w(a) = w(b)$. $\square$

Resilience (Amgoud and Ben-Naim 2017) states that the strength of arguments with an initial weight strictly between 0 and 1 can never reach the extreme values 0 and 1.

Proposition 11 (Resilience). *Let $A = (\mathcal{A}, w, \mathcal{R}, S)$ be a BAG that reaches an equilibrium s . If $a \in \mathcal{A}$ is such that $0 < w(a) < 1$, then $0 < s_a < 1$.*

Proof. The claim follows from Proposition 2. If $E_a \geq 0$, then equation (3) gives us $s_a = w(a) + (1 - w(a)) \cdot h(E_a)$ which is greater-or-equal to $w(a) > 0$ and strictly smaller than 1 because h is bounded from above by 1 and $w(a) < 1$. The case $E_a < 0$ is similar. $\square$

Franklin (Amgoud and Ben-Naim 2017) says that attackers and supporters with equal strength cancel their impact.

Proposition 12 (Franklin). *Let $A = (\mathcal{A}, w, \mathcal{R}, S)$ be a BAG that reaches an equilibrium s . If $a, b \in \mathcal{A}$ are such that $w(a) = w(b), \text{Att}_a = \text{Att}_b \cup \{x\}, \text{Sup}_a = \text{Sup}_b \cup \{y\}$ and $s_x = s_y$, then $s_a = s_b$.*

Proof. The claim follows from Proposition 2 because $E_a = E_b - s_x + s_y = E_b$. $\square$

Weakening (Amgoud and Ben-Naim 2017) says that the final strength of arguments that are strictly weaker supported than attacked must be smaller than the initial weight. Att^+ denotes again the attackers with strictly positive strength.

Proposition 13 (Weakening). *Let $A = (\mathcal{A}, w, \mathcal{R}, S)$ be a BAG that reaches an equilibrium s . Let $a \in \mathcal{A}$ with $w(a) > 0$. Suppose that $f : \text{Sup}_a \rightarrow \text{Att}_a$ is an injective function such that $s_x \leq s_{f(x)}$ for all $x \in \text{Sup}_a$ and $(\text{Att}_a^+ \setminus f(\text{Sup}_a) \neq \emptyset \text{ or there is an } x \in \text{Sup}_a \text{ such that } s_x < s_{f(x)})$. Then $s_a < w(a)$.*

Proof. The claim follows from Proposition 2 because the assumptions imply $E_a < 0$. $\square$

Strengthening (Amgoud and Ben-Naim 2017) is dual to *Weakening* and demands that for arguments that are strictly weaker attacked than supported, the strength must be greater than the initial weight. As before, Sup^+ denotes the supporters with strictly positive strength.

Proposition 14 (Strengthening). *Let $A = (\mathcal{A}, w, \mathcal{R}, S)$ be a BAG that reaches an equilibrium s . Let $a \in \mathcal{A}$ with $w(a) < 1$. Suppose that $f : \text{Att}_a \rightarrow \text{Sup}_a$ is an injective function such that $s_x \leq s_{f(x)}$ for all $x \in \text{Att}_a$ and $(\text{Sup}_a^+ \setminus f(\text{Att}_a) \neq \emptyset \text{ or there is an } x \in \text{Att}_a \text{ such that } s_x < s_{f(x)})$. Then $s_a > w(a)$.*

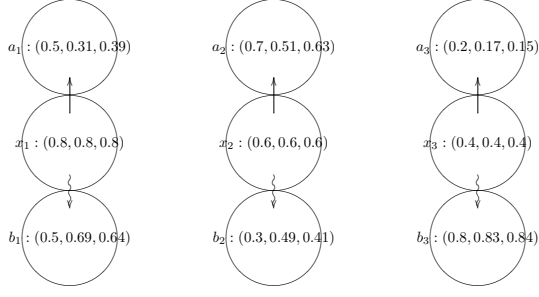


Figure 3: Duality Example. Nodes show (initial weight, strength under quadratic energy model, strength under Euler-based restricted semantics).

Proof. The claim follows from Proposition 2 because the assumptions imply $E_a > 0$. $\square$

This completes our discussion of axioms from (Amgoud and Ben-Naim 2017). The quadratic energy model satisfies all of them. We will now look at some differences between the quadratic energy model and the Euler-based restricted semantics from (Amgoud and Ben-Naim 2017).

As demonstrated in Section 2, the asymmetric nature of the Euler-based restricted semantics causes an unbalance between attack and support. As argued there, we may expect symmetrical changes for attacks and supports when the initial weight is 0.5. When the initial weight is greater (smaller) than 0.5, we cannot expect perfect symmetry because the weight is now closer to (more far away from) 1. However, suppose that a's and b's initial weights are shifted away from 0.5 by ϵ in different directions, a's attackers are equally strong as b's supporters and a's supporters are equally strong as b's attackers. Then if a's strength is shifted away by δ from its initial weight, b's strength should be shifted away by $-\delta$. We first prove that this is true for the quadratic energy model and illustrate the axiom afterwards.

Proposition 15 (Duality). *Let $\mathbf{A} = (\mathcal{A}, w, \mathcal{R}, \mathcal{S})$ be a BAG that reaches an equilibrium s . Let $a, b \in \mathcal{A}$ such that $w(a) = 0.5 + \epsilon$, $w(b) = 0.5 - \epsilon$ for some $\epsilon \in [0, 0.5]$. If there are bijections $f : \text{Att}_a \rightarrow \text{Sup}_b$, $g : \text{Sup}_a \rightarrow \text{Att}_b$ such that $s_x = s_{f(x)}$ and $s_y = s_{g(y)}$ for all $x \in \text{Att}_a, y \in \text{Sup}_a$, then $s_a - w(a) = w(b) - s_b$.*

Proof. The claim follows from Proposition 2. First note that the assumptions imply $E_a = -E_b$. If $E_a \geq 0$, we have $s_a - w(a) = (1 - (0.5 + \epsilon)) \cdot h(E_a) = (0.5 - \epsilon) \cdot h(-E_b) = -(s_b - w(b))$. We can argue similarly for $E_a < 0$. $\square$

We illustrate *Duality* in Figure 3. The subgraph with indices 1 shows a case similar to the one in Section 2. Under Euler-based semantics, x_1 decreases a_1 by 0.11, but increases b_1 by 0.14. Under the quadratic energy model, the effect of attack and support is 0.19 in both cases. The remaining two subgraphs illustrate more general cases. a_2 's and b_2 's initial weight is moved away by 0.2 from 0.5 in different directions. The attack on a_2 has the same impact as the support of b_2 . In this case, they both shift the strength

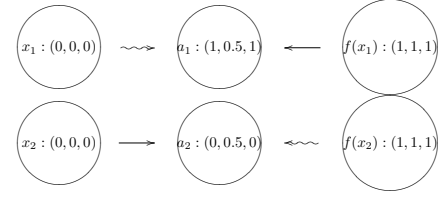


Figure 4: Weakening and Strengthening counterexample. Nodes show (initial weight, strength under quadratic energy model, strength under Euler-based restricted semantics).

away by 0.19 from the initial weight in different directions. Similarly, in the subgraph with indices 3, the shift is 0.03.

Another characteristic of the Euler-based semantic is that arguments with initial weight 0 or 1 can never change, no matter how strong their attackers and supporters are. This can be seen from the aggregation formula (1). If the initial weight is 0, both numerator and denominator in (1) are 1, so that the result will be 0. Similarly, if the initial weight is 1, the numerator will become 0, so that the result is necessarily 1. At first sight, it seems to be an interesting feature that the strength can be fixed to 0 or 1. However, the same effect can be achieved in every framework that satisfies *Stability* by just adding an argument without parents. Indeed, the parents of arguments with strength 0 or 1 become meaningless for the Euler-based semantics. For this reason, the Euler-based semantics satisfies the Weakening and the Strengthening axiom only if the initial weights 0 and 1 are excluded as we illustrate in Figure 4. Notice that according to Weakening (Strengthening), the strength of a_1 (a_2) should be strictly smaller (greater) than 1 (0). We also show the corresponding values for the quadratic energy model as before. The authors in (Amgoud and Ben-Naim 2017) introduced the notion of a *restricted semantics* that excludes certain initial weights in order to circumvent the problem.

5 Convergence and Computational Results

We showed that equilibria of the quadratic energy model satisfy several desirable properties. We will now turn to the question under what conditions the quadratic energy model actually reaches an equilibrium and how expensive computing the equilibrium is. We will show that for acyclic BAGs, we have the same guarantees like for discrete update procedures. Namely, the quadratic energy model is guaranteed to converge and equilibria can be computed in linear time. For general BAGs, we cannot give very interesting analytical guarantees but demonstrate empirically that the quadratic energy model still converges and does so quickly.

To begin with, we show that $\sigma^{\mathbf{A}}(t)$ is guaranteed to reach an equilibrium state for acyclic BAGs.

Proposition 16 (Convergence for Acyclic BAGs). *Let $\mathbf{A}$ be an acyclic BAG. Then $\sigma^{\mathbf{A}}$ reaches an equilibrium state s^* .*

Proof. Since $\mathbf{A}$ is acyclic, we can assume w.l.o.g. that the arguments are topologically ordered. That is, for every edge (i, j) (attack or support), we have $i < j$. We show that the derivative for the strength of each argument must vanish as $t \rightarrow \infty$. This then implies that the limit must exist.

Since 1 has no predecessors, the energy at 1 is 0 for all time. From equation (2), we get $\frac{d\sigma_1^A}{dt}(t) = w(1) - \sigma_1^A(t)$. Since, $\sigma_1^A(0) = w(1)$, the derivative is 0 for all time.

Assume that the derivative for the first $n - 1$ arguments vanishes. The energy at argument n can only be influenced by the arguments $1, \dots, n - 1$. Since the derivative at all these arguments vanishes, their strength converges to a limit. Hence, the energy at n will converge to a limit L . Assume $L > 0$. By continuity, the derivative at n will converge to

$$w(n) - \sigma_n^A(t) + (1 - w(n)) \cdot h(L).$$

Notice that if $\sigma_n^A(t)$ is greater (smaller) than the quantity $w(n) + (1 - w(n)) \cdot h(L)$, the derivative is negative (positive). Therefore, $\sigma_n^A(t)$ must eventually converge to $w(n) + (1 - w(n)) \cdot h(L)$ and the derivative will go to 0. We can argue similarly for the cases $L = 0$ and $L < 0$. $\square$

The proof of Proposition 16 actually shows that we can compute the equilibrium state in one pass through the acyclic graph analogous to the models in (Baroni et al. 2015; Rago et al. 2016; Amgoud and Ben-Naim 2017). We explain this in more detail in the following proposition.

Proposition 17 (Computing Equilibria in Acyclic BAGs). *Let A be an acyclic BAG. The equilibrium $s^* = \lim_{t \rightarrow \infty} \sigma^A(t)$ can be computed in linear time by the following procedure:*

1. Compute a topological ordering of the arguments.
2. Pick the next argument i in the order and set the energy at i to the value given in equation (3).
3. Repeat 2. until all strength values have been computed.

Proof. As shown in Proposition 16, when picking arguments according to a topological ordering of the arguments, the limit of all predecessors is known and the limit at the current argument is determined by these limits and equation (3). The procedure above just sets the strength of arguments to this limit. Recall that this limit is just the initial weight for arguments without parents. For other arguments, the topological ordering guarantees that the strength values of all predecessors has been set already, so that the energy can be computed. A topological ordering can be computed in time $O(|\mathcal{A}| + |\mathcal{R}| + |\mathcal{S}|)$ (Cormen et al. 2009). Afterwards, the strength values can be set in one pass through all arguments. Hence, the overall runtime is linear. $\square$

Example 2. One topological ordering for the BAG in Figure 1 is 3, 6, 7, 9, 4, 5, 0, 8, 1, 2. For 3, 6, 7, 9, the energy is 0 and we set the strength to their initial weight. Then the energy at 4, 5, 0, 8 is determined and we can set the strength values according to equation (3). Afterwards, the energy at 1, 2 is determined and the remaining strength values can be set. The reader can check that this procedure yields the same result as the continuous process illustrated in Figure 2 (up to rounding errors). Every other order that guarantees that we evaluate nodes before their children will work as well. For example, since 9 has no parents, we can just set $s_9 = w(9) = 0.99$. We are then ready to compute s_8 because 9 is the only parent of 8. We have $E_8 = 0.99$ and $w(8) = 0.1$,

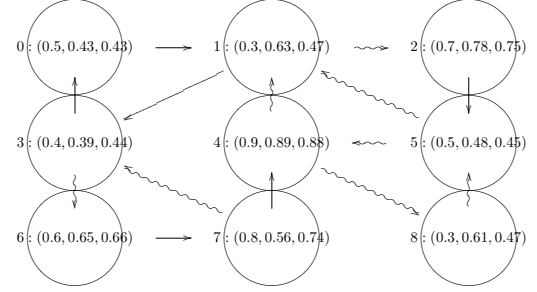


Figure 5: Cyclic BAG. Nodes show (initial weight, strength under quadratic energy model, strength under continuous Euler-based model).

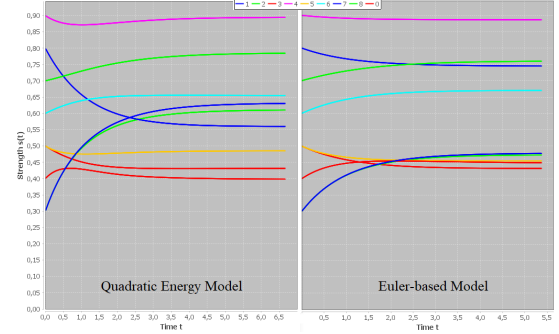


Figure 6: Long-term behaviour of quadratic energy model vs. continuous Euler-based model for BAG in Figure 5.

so the strength at 8 must be $0.1 + 0.9 \cdot \frac{0.99^2}{1+0.99^2} \approx 0.5455$. We can check in Figure 2 that this is indeed the value that argument 8's strength converges to.

Let us now look at cyclic BAGs. Figure 5 shows an example. For every node, the first value shows again the initial weight and the second value the final strength under the quadratic energy model. The third value corresponds to a continuized version of the Euler-based semantics that we will explain in Section 6. Please ignore this entry for now. The long-term behaviour of the quadratic energy models is plotted in the left subfigure of Figure 6. Note that our guarantees about axiomatic properties in equilibria still hold because they are not based on assumptions about acyclicity. The remaining question is not if equilibria satisfy our guarantees, but if we can guarantee that the model always converges to an equilibrium for acyclic BAGs.

Unfortunately, analyzing the long-term behaviour of σ^A is complicated for cyclic BAGs. Currently, we can only give convergence guarantees for some simple special cases. For instance, if the BAG contains only support relations, the strength at each argument can only increase. Since the strength is bounded by 1, σ^A must eventually converge. Another simple case is obtained if we take an acyclic BAG and 'replace' a node without parents with a cycle that is already known to converge. We can then argue as before that the strength values must converge because the strength values in the cycle will eventually converge. Admittedly, these special cases are probably not very widely applicable. How-

ever, such an inductive proof strategy may be successful to give more interesting convergence guarantees in the future.

Experimentally, the quadratic energy model still converges for cyclic BAGs and does so quickly. For the BAG shown in Figure 5, σ^A converged until time 7. The number of computational steps depends on the step size that we choose for our numerical approximation algorithm. Implementations in numerical libraries often adapt the step size automatically. However, since the absolute value of the derivatives for the quadratic energy model can never become larger than 1, using constant step size 0.01 for RK4 seems safe. This means that we have to perform 700 evaluations of the Jacobian matrix from time 0 to time 7. RK4 and other sophisticated algorithms evaluate the Jacobian matrix at multiple points to improve approximation guarantees, so that we actually have a constant multiple of 700. In general, if we use step size δ and σ^A converges at time T , we will need $O(\frac{T}{\delta})$ evaluations. The evaluation of the Jacobian matrix comes down to iterating over all arguments and all their attackers and supporters. Hence, evaluation can be done in linear time. As we show in Section 6, we can sometimes create similar dynamical systems by continuizing discrete update procedures. We therefore generalize our observations in the following proposition. f_D generalizes the right-hand side of equation (2) and describes how to adapt the strength of arguments based on their initial weight and the current strength of attackers and supporters.

Proposition 18 (Runtime for Cyclic BAGs). *Consider a BAG A and a solution $\sigma_D^A : \mathbb{R}_0^+ \rightarrow \mathbb{R}^n$ of the system of differential equations $\frac{ds}{dt} = f_D(w, s)$. If f_D can be computed in time $O(|A| + |\mathcal{R}| + |\mathcal{S}|)$, then approximating σ_D^A with step size δ until time T can be done in time $O(\frac{(|A| + |\mathcal{R}| + |\mathcal{S}|) \cdot T}{\delta})$.*

In general, the point of convergence T can depend on the size of the BAG and the size and complexity of cycles. In order to evaluate the runtime empirically, I created random cyclic BAGs as follows: The number of arguments n was increased from 100 to 3,000 in steps of size 100. For each n , 100 random graphs were generated. To this end, cycles of random size between 3 and $\sqrt{n}$ were added. These cycles contained fresh arguments and contained either only support, only attack or uniformly distributed edges. The cycle type was chosen uniformly at random. Such cycles were generated independently until the number of arguments reached n . The size of the last cycle was chosen to obtain n arguments overall (so it can also be a self-loop of size 1). Subsequently, arguments $1, \dots, \frac{n}{2}$ and $\frac{n}{2} + 1, \dots, n$ were connected to two large cycles of length $\frac{n}{2}$ with uniformly distributed edges. Finally, for each argument a , with probability $\frac{1}{2}$, between 1 and $\sqrt{n}$ (chosen uniformly) mixed edges (chosen uniformly) were added from a to a uniformly chosen argument (self loops are possible). The random generator made sure that for each pair i, j there is at most one edge (either attack or support) from i to j . The initial weight for all arguments was chosen uniformly between 0 and 1. The expected number of edges is in the order of $\Theta(n^{\frac{3}{2}})$ and the probability that generated graphs form a single strongly connected component is almost 1. I checked some random in-

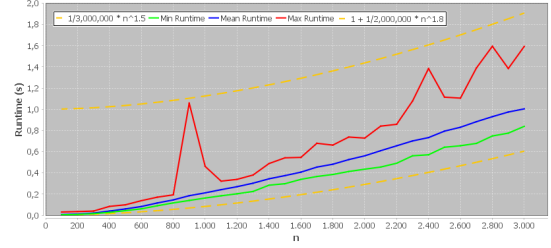


Figure 7: Runtime results for computing equilibria. Minimum, mean and maximum runtime are taken over 100 random cyclic BAGs with n arguments and $\Theta(n^{\frac{3}{2}})$ edges.

stances in order to make sure that significant changes in the strength of arguments occurred. For instance, for a random instance with $n = 1,000$, the average absolute change in arguments' strength was 0.26 and ranged from 0 to 0.9. The BAG contained about 4,500 attacks and 4,500 supports. An archive containing all BAGs can be downloaded from the link given in the footnote¹.

I implemented RK4 in Java with fixed step size $\delta = 0.01$. The algorithm terminates when all components in the Jacobian matrix are smaller than $\epsilon = 10^{-4}$. The experiments ran on a single core of an Intel Core i7-4770 with 8 GB memory and Windows 10 Home operating system. Figure 7 shows, for each n , minimum, mean and maximum runtime for the 100 trials in seconds and function values of $f(n) = 1 + \frac{1}{3,000,000} \cdot n^{1.5}$ and $g(n) = 1 + \frac{1}{2,000,000} \cdot n^{1.8}$ as lower and upper bounds. Since the expected number of edges is in the order of $\Theta(n^{1.5})$, the experiments indicate that the quadratic energy system converges slightly super-linearly, but clearly subquadratically for cyclic BAGs. An implementation with adaptive step-size will probably further improve the runtime. For none of the 3,000 randomly generated cyclic BAGs oscillations occurred. Of course, this is not proof that oscillations cannot occur. In particular, there may be a systematic reason for oscillations in cyclic BAGs that the experiments are missing. However, the experiments give empirical evidence that the quadratic energy model converges for many cyclic BAGs. Note also that we can recognize oscillations in parts of the model automatically by checking if derivatives change sign repeatedly and manually by plotting the behaviour over time like in Figure 2 and 6.

6 Continuizing Discrete Models

Finally, we will close the circle between continuous dynamical systems and discrete iterative procedures like in (Baroni et al. 2015; Rago et al. 2016; Amgoud and Ben-Naim 2017). Roughly speaking, Proposition 17 says that we can transform the continuous quadratic energy model into a discrete update procedure for acyclic graphs. Sometimes, we can also do the converse and transform discrete update procedures to continuous dynamical systems with guarantees similar to those for the quadratic energy model.

¹https://www.researchgate.net/publication/326557254_Weighted_Bipolar_Argumentation_Benchmark_KR2018

Proposition 19 (Continuizing Iterative Schemes). *Consider an iterative scheme $\mathcal{I}$ that defines the strength values for acyclic BAGs by letting*

$$s_i = f_{\mathcal{I}}(w(i), \{s_j \mid j \in \text{Att}_i\}, \{s_j \mid j \in \text{Sup}_i\}),$$

where $f_{\mathcal{I}}$ is a function that depends on the initial weight and the strength of attackers and supporters and the strength values s_i are computed in topological order.

1. If $f_{\mathcal{I}}$ is continuously differentiable with respect to all involved strength values, then for all BAGs A , the system

$$\frac{ds_i}{dt} = f_{\mathcal{I}}(w(i), \{s_j \mid j \in \text{Att}_i\}, \{s_j \mid j \in \text{Sup}_i\}) - s_i$$

with initial conditions $s_i(0) = w(i)$ for $i = 1, \dots, n$ has a unique solution $\sigma_{\mathcal{I}}^A : \mathbb{R}_0^+ \rightarrow \mathbb{R}^n$.

2. If $\sigma_{\mathcal{I}}^A$ reaches an equilibrium state $s^* = \lim_{t \rightarrow \infty} \sigma_{\mathcal{I}}^A(t)$, then $s_i^* = f_{\mathcal{I}}(w(i), \{s_j^* \mid j \in \text{Att}_i\}, \{s_j^* \mid j \in \text{Sup}_i\})$.
3. $\sigma_{\mathcal{I}}^A$ reaches an equilibrium state whenever A is acyclic.

Proof. Existence and uniqueness of the solution follow analogously to Proposition 1. Analogously to Proposition 2, one can show that in the limit $s_i = f_{\mathcal{I}}(w(i), \{s_j \mid j \in \text{Att}_i\}, \{s_j \mid j \in \text{Sup}_i\})$. Convergence for acyclic BAGs can be proved analogously to Proposition 16. $\square$

Let us think about what continuizing an iterative scheme $\mathcal{I}$ means. Suppose we apply Euler's methods for solving initial value problems to this system. Then we update each strength value s_i with $s_i \leftarrow s_i + \delta \cdot \frac{ds_i}{dt}$ in every iteration. If we let $\delta = 1$, the update is just

$$s_i + 1 \cdot (f_{\mathcal{I}}(w(i), \{s_j \mid j \in \text{Att}_i\}, \{s_j \mid j \in \text{Sup}_i\}) - s_i) \\ = f_{\mathcal{I}}(w(i), \{s_j \mid j \in \text{Att}_i\}, \{s_j \mid j \in \text{Sup}_i\}).$$

That is, we just update all strength values simultaneously with respect to the iterative update formula $f_{\mathcal{I}}$. Of course, it is not a very deep idea to just apply the discrete update scheme $\mathcal{I}$ simultaneously and to see what happens. However, in the presence of cycles, this may lead to oscillations even when the continuous model $\sigma_{\mathcal{I}}^A$ converges. Intuitively, this is because a large step size like $\delta = 1$ can let us jump from the graph of the true solution $\sigma_{\mathcal{I}}^A$ to the graph of another solution for different initial conditions. By choosing a smaller step size, we can avoid these jumps and make the procedure more stable. This is basically what we are doing when continuizing $\mathcal{I}$.

The iteration scheme that corresponds to the Euler-based restricted semantics from (Amgoud and Ben-Naim 2017) satisfies the assumptions in Proposition 19. f is given by equation (1) and the system of differential equations is

$$\frac{ds_j}{dt} = \left(1 - \frac{1 - w(j)^2}{1 + w(j) \cdot \exp(E)}\right) - s_j, \quad j \in \mathcal{A}. \quad (4)$$

The continuous Euler-based model also satisfies the assumptions for our runtime analysis in Proposition 18. The j -th component of $f_D(w, s)$ is defined by equation (4).

In Figure 8, we show the long-term behaviour of the continuous Euler-based model for the acyclic BAG in Figure 1.

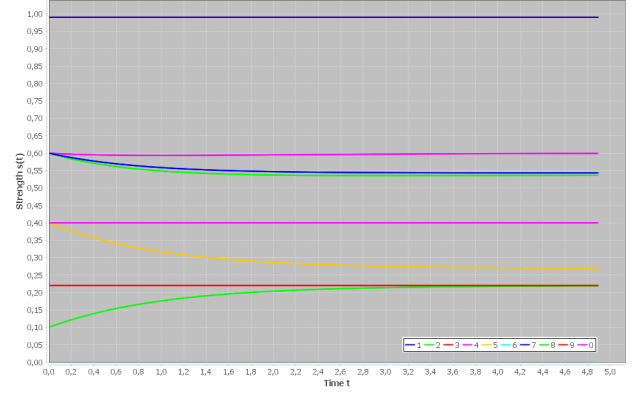


Figure 8: Long-term behaviour of continuous Euler-based model for BAG in Figure 1.

Note that it does indeed converge to the strength values defined by the Euler-based semantics. Continuizing a discrete update scheme has no computational advantages for acyclic BAGs. However, the option of plotting the continuous evolution of the strength values may be interesting to explain the final strength values to users.

The right subfigure in Figure 6 shows the long-term behaviour of the continuous Euler-based model for the BAG in Figure 5. Notice that the quadratic energy model converges at about time 6.5, the continuous Euler-based model at time 5.5 already. This is likely to happen in general because the definition of the Euler-based semantics creates relatively small derivatives. As a consequence, the final strength values under Euler-based semantics will also stay closer to the initial weights.

7 Conclusions

We introduced the quadratic energy model that satisfies all axioms for weighted bipolar argumentation proposed in (Amgoud and Ben-Naim 2017). It also satisfies the *Duality* property that guarantees a symmetric impact of attack and support relations and distinguishes it from the Euler-based semantics from (Amgoud and Ben-Naim 2017). For acyclic BAGs, equilibria can be found similar to the approaches in (Baroni et al. 2015; Rago et al. 2016; Amgoud and Ben-Naim 2017) by a single pass through the graph. For cyclic BAGs, we can currently only prove convergence guarantees for very restricted cases like BAGs that contain only support relations. However, experiments indicate that the quadratic energy model also converges for very complex cyclic BAGs and does so quickly even for BAGs with thousands of arguments and then thousands of attacks and supports. Of course, the quadratic energy model may have its own flaws that I am missing. However, as nicely put in (Rago et al. 2016), different behaviour can be desirable in different contexts, so that our goal should be the construction of a portfolio of suitable algorithms rather than the quest for a best one. In this spirit, I understand the quadratic energy model as a contribution to such a portfolio with its own advantages and disadvantages. In a similar spirit, I do not claim that

continuous dynamical systems are the ultimate solution for dealing with cyclic BAGs. However, I think that they are a well-suited tool for this area. Furthermore, the extensive literature on dynamical systems may be an interesting source to enhance our understanding of the convergence behaviour of both discrete and continuous algorithms for cyclic BAGs.

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²<http://www.jfree.org/jfreechart/>